

lean4ml

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Chapter 1

General Definitions and Basic Properties

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, $s \subseteq E$, and $L \in \mathbb{R}_{\geq 0}$ (Lean: `NNReal`).

Many of the properties presented here are well-known and we don't give an explicit proof. Still, these were added to our Lean project to simplify the proofs of more complex results.

1.1 L -smoothness and others

Definition 1 (L -smoothness on a set). We say that f is L -smooth on s if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in s.$$

Definition 2 (Global L -smoothness). A function is globally L -smooth if it is L -smooth on all of E .

Lemma 3 (Global smoothness is smoothness on `Set.univ`). *In Lean:*

$$L\text{Smooth} f L \iff L\text{SmoothOn } f L \text{ Set.univ}.$$

Lemma 4 (Monotonicity under set inclusion). *If $T \subseteq S$ and f is L -smooth on S , then f is L -smooth on T .*

Definition 5 (Hessian). The Hessian is the Fréchet derivative of the gradient map:

$$\nabla^2 f(x) := D(\nabla f)(x).$$

Definition 6 (Globally μ -PL function). We say that f is globally μ -PL if for all x we have

$$\|\nabla f(x)\|^2 \geq 2\mu(f(x) - \inf f).$$

1.2 Line Map and its properties

Definition 7 (Line map). The canonical line map through x in direction p is $t \mapsto x + tp$.

Lemma 8 (Continuity of $t \mapsto (Df(x + tp))p$). *If f is C^1 , then $t \mapsto (Df(x + tp))p$ is continuous.*

Lemma 9 (Continuity of $t \mapsto \langle \nabla f(x+tp), p \rangle$). *If f is C^1 , then $t \mapsto \langle \nabla f(x+tp), p \rangle$ is continuous.*

Proof. In Lean, simply rewrite $(Df(x+tp))p$ as $\langle \nabla f(x+tp), p \rangle$ using `HasGradientAt.fderiv_apply`. $\square$

Lemma 10 (Integrability of the line integrand). *The map $t \mapsto \langle \nabla f(x+tp), p \rangle$ is integrable on any interval (by continuity).*

Lemma 11 (Chain rule along a line). *If f has Fréchet derivative at $x+tp$, then $\tau \mapsto f(x+\tau p)$ has derivative $(Df(x+tp))p$ at $\tau = t$.*

Lemma 12 (Convex segment stays in a convex set). *If s is convex and $x, y \in s$, then $x+t(y-x) \in s$ for $t \in [0, 1]$.*

Lemma 13 (Scaling of the Hessian quadratic form).

$$\langle \nabla^2 f(x)(\alpha p), \alpha p \rangle = \alpha^2 \langle \nabla^2 f(x)p, p \rangle.$$

Lemma 14 (Hessian continuity along a line). *If $f \in C^2$, then $t \mapsto \nabla^2 f(x+tp)p$ is continuous.*

1.3 Other intermediary results

Lemma 15 (Derivative operator-norm bound from Lipschitz). *If g is L -Lipschitz on a set s containing a neighborhood of x and g is differentiable at x , then $\|Dg(x)\| \leq L$.*

Proof. Simply by noting that for $y \neq x$ in the neighborhood, $\frac{|f(x)-f(y)|}{\|x-y\|} \leq L$, from where the limit definition of the derivative norm gives $\|Dg(x)\| \leq L$. $\square$

Lemma 16 (Quadratic form controlled by operator norm). *For $A : E \rightarrow \mathcal{L}[\mathbb{R}]E$ and $p \in E$:*

$$\langle Ap, p \rangle \leq \|A\| \|p\|^2.$$

Lemma 17 (HasGradient everywhere for C^1). *If $f \in C^1(E)$, then for every $z \in E$ the gradient exists.*

Lemma 18 (Norm Lipschitz bound for the gradient). *If f is L -smooth on s and $x, y \in s$, then $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.*

Lemma 19 (Segment norm identity). *For $\gamma \geq 0$, $\|(x + \gamma(y - x)) - x\| = \gamma\|y - x\|$.*

Lemma 20 (Integral split-off constant term in an inner product). *Lean lemma used to rewrite $\int \langle g(t) - c, w \rangle = \int \langle g(t), w \rangle - \int \langle c, w \rangle$.*

Lemma 21 (Monotonicity of interval integrals (continuous case)). *If $\varphi \leq \psi$ on $[a, b]$ with $a \leq b$ and both are continuous, then $\int_a^b \varphi \leq \int_a^b \psi$.*

Lemma 22 (Inner-product linearity rewrite). $\langle u, w \rangle - \langle v, w \rangle = \langle u - v, w \rangle$.

Chapter 2

Taylor's Theorem

Theorem 23 (First-order Taylor formula along a line (vector-valued)). *If $f : E \rightarrow F$ is C^1 , then*

$$f(x + p) - f(x) = \int_0^1 (Df(x + tp))p \, dt.$$

Proof. Let $g(t) := f(x + tp)$ ($t \in \mathbb{R}$). We apply the fundamental theorem of calculus to g on $[0, 1]$.

For each t , since $f \in C^1$, f is Fréchet differentiable at $x + tp$, with derivative $Df(x + tp)$. By the chain rule along a line (Lemma 11), $g'(t) = (Df(x + tp))p$. Hence g has the required derivative at every point of $\text{ulcc}(0, 1)$.

Next, the integrand $t \mapsto (Df(x + tp))p$ is interval-integrable on $[0, 1]$ by continuity (Lemma 8).

Therefore, by `intervalIntegral.interval_eq_sub_of_hasDerivAt`,

$$\int_0^1 (Df(x + tp))p \, dt = g(1) - g(0).$$

Expanding $g(1) = f(x + p)$ and $g(0) = f(x)$ gives

$$f(x + p) - f(x) = \int_0^1 (Df(x + tp))p \, dt.$$

□

Theorem 24 (First-order Taylor formula along a line). *If $f \in C^1$, then*

$$f(x + p) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle \, dt.$$

Proof. Direct from Theorem 23. □

Theorem 25 (Second-order Taylor formula for the gradient). *If $f \in C^2$, then*

$$\nabla f(x + p) - \nabla f(x) = \int_0^1 \nabla^2 f(x + tp) p \, dt.$$

Proof. Direct from Theorem 23 applied to $\nabla f \in C^1$. □

Lemma 26 (Pointwise Lipschitz-gradient bound). *For $f \in C^1$ that is L -smooth, for $\gamma \geq 0$ and the segment point in s ,*

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x \rangle \leq L\gamma \|y - x\|^2.$$

Proof. First, derive a norm bound from L -smoothness. Since $x \in s$ and $x + \gamma(y - x) \in s$, Lemma 18 gives

$$\|\nabla f(x + \gamma(y - x)) - \nabla f(x)\| \leq L\|(x + \gamma(y - x)) - x\|.$$

Using Lemma 19 (with $\gamma \geq 0$), $\|(x + \gamma(y - x)) - x\| = \gamma\|y - x\|$, hence $\|\nabla f(x + \gamma(y - x)) - \nabla f(x)\| \leq L\gamma\|y - x\|$.

Second, apply Cauchy-Schwarz:

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x \rangle \leq \|\nabla f(x + \gamma(y - x)) - \nabla f(x)\| \|y - x\|.$$

Plug the previous bound into the right-hand side to get

$$\langle \nabla f(x + \gamma(y - x)) - \nabla f(x), y - x \rangle \leq (L\gamma\|y - x\|)\|y - x\| = L\gamma\|y - x\|^2,$$

which is exactly the claim. $\square$

Lemma 27 (Non-parametric corollary ($\gamma = 1$)). *In particular, if $x, y \in s$ then $\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq L\|y - x\|^2$.*

Theorem 28 (Quadratic upper bound from smoothness on a convex set). *If $f \in C^1$, f is L -smooth on convex s , and $x, y \in s$, then*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. Let $p := y - x$.

1. Apply Theorem 24 to obtain $f(y) - f(x) = \int_0^1 \langle \nabla f(x + tp), p \rangle dt$.
2. Rewrite $f(y) - f(x) - \langle \nabla f(x), p \rangle$ as an integral of $\langle \nabla f(x + tp) - \nabla f(x), p \rangle$ using Lemma 20.
3. Use convexity via Lemma 12 to ensure $x + tp \in s$ for $t \in [0, 1]$.
4. Bound the integrand pointwise using Lemma 26 with $\gamma = t$, then integrate the bound with Lemma 21.
5. Evaluate $\int_0^1 Lt dt = L/2$ and conclude.

$\square$

Theorem 29 (Descent Lemma for Gradient Descent). *If $f \in C^1$, f is L -smooth on a convex set s , $x \in s$, $\alpha \geq 0$, and $x - \alpha \nabla f(x) \in s$, then*

$$f(x - \alpha \nabla f(x)) \leq f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|^2.$$

Proof. Let $y := x - \alpha \nabla f(x)$ and $p := y - x = -\alpha \nabla f(x)$.

Step 1: Compute the inner product.

$$\langle \nabla f(x), p \rangle = \langle \nabla f(x), -\alpha \nabla f(x) \rangle = -\alpha \|\nabla f(x)\|^2.$$

Step 2: Compute the norm squared of p using $\alpha \geq 0$ to simplify $|\alpha| = \alpha$.

$$\|p\|^2 = \|- \alpha \nabla f(x)\|^2 = (\|\alpha \nabla f(x)\|)^2 = \alpha^2 \|\nabla f(x)\|^2.$$

Step 3: Apply Theorem 28 with the points x and $y = x + p$:

$$f(y) \leq f(x) + \langle \nabla f(x), p \rangle + \frac{L}{2} \|p\|^2.$$

Step 4: Substitute the expressions from Steps 1 and 2:

$$\begin{aligned} f(y) &\leq f(x) + (-\alpha \|\nabla f(x)\|^2) + \frac{L}{2} (\alpha^2 \|\nabla f(x)\|^2) \\ &= f(x) - \alpha \|\nabla f(x)\|^2 + \frac{L\alpha^2}{2} \|\nabla f(x)\|^2 \\ &= f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|^2. \end{aligned}$$

This completes the proof. □

Chapter 3

Hessian characterization of L-Smooth functions

Theorem 30 (Hessian operator-norm bound $\Rightarrow$ global smoothness). *If ∇f is differentiable everywhere with derivative $\nabla^2 f(x)$, and $\|\nabla^2 f(x)\| \leq L$ for all x , then f is globally L -smooth.*

Proof. We need to show ∇f is L -Lipschitz globally.

Since ∇f is Fréchet differentiable at every point x with derivative $\nabla^2 f(x)$ (by hypothesis), and we have the operator-norm bound $\|\nabla^2 f(x)\| \leq L$ for all x , Mathlib's `lipschitzWith_of_nnorm_fderiv_le` directly gives that ∇f is L -Lipschitz on all of E . By definition, this means f is globally L -smooth. $\square$

Theorem 31 (Global smoothness $\Rightarrow$ Hessian quadratic-form upper bound). *If f is globally L -smooth, then $\langle \nabla^2 f(x)p, p \rangle \leq L\|p\|^2$ for all x, p .*

Proof. Fix x and p . By global L -smoothness, ∇f is L -Lipschitz on all of E .

Apply Lemma 15 to the gradient map: since ∇f is L -Lipschitz on a neighborhood of x (in fact, globally) and is Fréchet differentiable at x with derivative $\nabla^2 f(x)$, we obtain

$$\|\nabla^2 f(x)\| \leq L.$$

By Lemma 16 (operator norm controls the quadratic form),

$$\langle \nabla^2 f(x)p, p \rangle \leq \|\nabla^2 f(x)\| \cdot \|p\|^2 \leq L\|p\|^2,$$

as required. $\square$

Chapter 4

Necessary Conditions

Theorem 32 (Local minimum $\Rightarrow$ gradient vanishes). *If $f \in C^1$ and x is a local minimizer (Lean: `IsLocalMin f x`), then $\nabla f(x) = 0$.*

Proof. By Lemma 17, the gradient $\nabla f(x)$ exists at the local minimizer x .

Since x minimizes f locally, the Fréchet derivative of f at x must vanish when applied to any direction. The Fréchet derivative of f at x is the inner product with $\nabla f(x)$, so we have

$$\langle \nabla f(x), v \rangle = 0 \quad \text{for all } v \in E.$$

Setting $v = \nabla f(x)$ gives $\|\nabla f(x)\|^2 = 0$, hence $\nabla f(x) = 0$. $\square$

Lemma 33 (Gradient as integral of Hessian along a line). *If $f \in C^2$ and $\nabla f(x) = 0$, then $\nabla f(x + tp) = \int_0^t \nabla^2 f(x + sp) p \, ds$.*

Lemma 34 (Exact second-order Taylor expansion (double integral)). *If $f \in C^2$ and $\nabla f(x) = 0$, then*

$$f(x + p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x + sp) p, p \rangle \, ds \, dt.$$

Theorem 35 (Local minimum $\Rightarrow$ Hessian PSD). *If $f \in C^2$ and x is a local minimizer, then $\langle \nabla^2 f(x) p, p \rangle \geq 0$ for all p .*

Proof. We proceed by contradiction. First, by Theorem 32, we may assume $\nabla f(x) = 0$.

Suppose $\langle \nabla^2 f(x) p, p \rangle < 0$ for some $p \neq 0$. Define $H(u) := \langle \nabla^2 f(x + up) p, p \rangle$. Then $H(0) = \langle \nabla^2 f(x) p, p \rangle < 0$. By continuity of the Hessian (Lemma 14), there exists $\delta > 0$ such that $H(u) < H(0)/2 < 0$ for all $u \in (-\delta, \delta)$.

But x is a local minimizer, so $f(x + \alpha p) \geq f(x)$ for small $\alpha > 0$. By the exact second-order Taylor formula (Lemma 34),

$$f(x + \alpha p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x + s\alpha p) p, p \rangle \, ds \, dt = \alpha^2 \int_0^1 \int_0^t \langle \nabla^2 f(x + s\alpha p) p, p \rangle \, ds \, dt.$$

For small $\alpha > 0$, all points $x + s\alpha p$ lie in the ball where $H < 0$, so both integrals are negative, making $f(x + \alpha p) - f(x) < 0$, contradicting local minimality. $\square$

Chapter 5

Sufficient Conditions

Definition 36 (Strict local minimizer). Lean definition: $\exists U \in \mathcal{N}(x), \forall y \in U, y \neq x \rightarrow f(x) < f(y)$.

Lemma 37 (Taylor lower bound from uniform Hessian lower bound). *Under a uniform bound $\varepsilon\|p\|^2 \leq \langle \nabla^2 f(y)p, p \rangle$ on a ball around x and assuming $\nabla f(x) = 0$, Lean proves $(\varepsilon/2)\|p\|^2 \leq f(x+p) - f(x)$ for $\|p\| < \rho$ by integrating the pointwise bound inside the double-integral remainder formula.*

Proof. Start from the exact Taylor formula (Lemma 34):

$$f(x+p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x+sp)p, p \rangle ds dt.$$

By convexity (or simply by norm estimates), the point $x+sp$ stays within the ball $B(x, \rho)$ for all $s \in [0, 1]$ when $\|p\| < \rho$. Within this ball, the Hessian quadratic form is bounded below: $\langle \nabla^2 f(y)p, p \rangle \geq \varepsilon\|p\|^2$.

Therefore,

$$f(x+p) - f(x) = \int_0^1 \int_0^t \langle \nabla^2 f(x+sp)p, p \rangle ds dt \geq \int_0^1 \int_0^t \varepsilon\|p\|^2 ds dt = \varepsilon\|p\|^2 \int_0^1 t dt = \frac{\varepsilon}{2}\|p\|^2,$$

as claimed. $\square$

Lemma 38 (Coercivity from positive definiteness (finite-dimensional)). *In finite dimensions, a positive definite continuous linear map H satisfies $\exists \varepsilon > 0, \forall p, \varepsilon\|p\|^2 \leq \langle Hp, p \rangle$. Lean uses compactness of the sphere and `IsCompact.exists_isLeast`.*

Lemma 39 (Continuity of Hessian map). *If $f \in C^2$, then $y \mapsto \nabla^2 f(y)$ is continuous (Lean: `continuous_fderiv` applied to ∇f).*

Lemma 40 (Positive definite Hessian $\Rightarrow$ uniform lower bound on a ball). *In finite dimensions, if $\nabla^2 f(x)$ is positive definite, then there exist $\varepsilon > 0$ and $\rho > 0$ such that $\varepsilon\|p\|^2 \leq \langle \nabla^2 f(y)p, p \rangle$ for all $y \in B(x, \rho)$ and all p .*

Proof. Since $\nabla^2 f(x)$ is positive definite, Lemma 38 provides $\varepsilon_0 > 0$ such that

$$\varepsilon_0\|p\|^2 \leq \langle \nabla^2 f(x)p, p \rangle \quad \text{for all } p.$$

By Lemma 39, the map $y \mapsto \nabla^2 f(y)$ is continuous. Therefore, we can find $\rho > 0$ such that

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| < \frac{\varepsilon_0}{2} \quad \text{for all } y \in B(x, \rho).$$

For $y \in B(x, \rho)$ and any p , the operator norm inequality gives

$$|\langle (\nabla^2 f(y) - \nabla^2 f(x))p, p \rangle| \leq \|\nabla^2 f(y) - \nabla^2 f(x)\| \|p\|^2 < \frac{\varepsilon_0}{2} \|p\|^2.$$

Therefore,

$$\langle \nabla^2 f(y)p, p \rangle = \langle \nabla^2 f(x)p, p \rangle + \langle (\nabla^2 f(y) - \nabla^2 f(x))p, p \rangle > \varepsilon_0 \|p\|^2 - \frac{\varepsilon_0}{2} \|p\|^2 = \frac{\varepsilon_0}{2} \|p\|^2,$$

which completes the proof. $\square$

Theorem 41 (Strict local minimizer from uniform Hessian lower bound). *If $f \in C^2$, $\nabla f(x) = 0$, and the Hessian has a uniform positive lower bound on a ball around x , then x is a strict local minimizer.*

Proof. Let $U = B(x, \rho)$. For any $y \in U$ with $y \neq x$, set $p = y - x$. Then $\|p\| < \rho$ and $f(y) - f(x) = f(x + p) - f(x)$.

By Lemma 37 (applied to the uniform lower bound on $\nabla^2 f$ over the ball around x), we have

$$f(x + p) - f(x) \geq \frac{\varepsilon}{2} \|p\|^2.$$

Since $p \neq 0$, we have $\|p\|^2 > 0$, so $f(y) > f(x)$. This holds for all $y \in U$ with $y \neq x$, showing that x is a strict local minimizer. $\square$

Theorem 42 (Second-order sufficient condition (finite-dimensional)). *If $f \in C^2$, $\nabla f(x) = 0$, and $\nabla^2 f(x)$ is positive definite, then x is a strict local minimizer.*

Proof. By hypothesis, $f \in C^2$ and $\nabla f(x) = 0$. The Hessian $\nabla^2 f(x)$ is positive definite.

Lemma 40 guarantees that there exist $\varepsilon > 0$ and $\rho > 0$ such that

$$\varepsilon \|p\|^2 \leq \langle \nabla^2 f(y)p, p \rangle \quad \text{for all } y \in B(x, \rho) \text{ and all } p.$$

With this uniform lower bound in place, Theorem 41 applies immediately to conclude that x is a strict local minimizer. $\square$

Chapter 6

GradientFlow

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, and ∇f denotes the (Fréchet) gradient in the Hilbert space structure.

Definition 43 (Gradient flow on a bounded interval). A *bounded-time* gradient flow is a certificate that the ODE

$$\dot{x}(t) = -\nabla f(x(t)), \quad x(0) = x_0$$

has a solution on $[0, T]$. In Lean, a `GradientFlow f x T` is a structure with fields:

- `trajectory : ℝ → E`,
- `init : trajectory 0 = x`,
- `hasDerivAt : t ↦ Icc 0 T, HasDerivAt trajectory (-(gradient f (trajectory t))) t`.

Lemma 44 (Initial value). For $\varphi : \text{GradientFlow } f \ x \ T$, we have $\varphi(0) = x_0$.

Lemma 45 (Differentiability of the trajectory on $[0, T]$). For $t \in [0, T]$, the map $t \mapsto \varphi(t)$ is differentiable at t . (Lean: `DifferentiableAt .trajectory t`.)

Lemma 46 (Derivative equals negative gradient). For $t \in [0, T]$,

$$\varphi'(t) = -\nabla f(\varphi(t)).$$

(Lean: `deriv .trajectory t = -(gradient f (.trajectory t))`.)

Lemma 47 (Energy derivative along the flow). If $f \in C^1$ and φ is a gradient flow, then for $t \in [0, T]$

$$\frac{d}{dt} f(\varphi(t)) = -\|\nabla f(\varphi(t))\|^2.$$

Proof. By the chain rule, the derivative of the composition $f \circ \varphi$ at time t is

$$\frac{d}{dt} f(\varphi(t)) = Df(\varphi(t)) \cdot \varphi'(t) = \langle \nabla f(\varphi(t)), \varphi'(t) \rangle.$$

Substituting the ODE $\varphi'(t) = -\nabla f(\varphi(t))$ gives

$$\frac{d}{dt} f(\varphi(t)) = \langle \nabla f(\varphi(t)), -\nabla f(\varphi(t)) \rangle = -\|\nabla f(\varphi(t))\|^2,$$

as claimed. □

Lemma 48 (Energy is non-increasing). *If $0 \leq r \leq s \leq T$, then*

$$f(\varphi(s)) \leq f(\varphi(r)).$$

Proof. Suppose for contradiction that $f(\varphi(s)) > f(\varphi(r))$. By the mean value theorem, there exists $t \in (r, s)$ such that

$$\varphi'(t) = \frac{f(\varphi(s)) - f(\varphi(r))}{s - r} > 0.$$

But by Lemma 47, we have

$$\varphi'(t) = -\|\nabla f(\varphi(t))\|^2 \leq 0,$$

a contradiction. Thus $f(\varphi(s)) \leq f(\varphi(r))$. $\square$

Definition 49 (Piecewise trajectory). The piecewise trajectory that follows φ_1 up to time T and then φ_2 (shifted) is

$$\Phi(t) := \begin{cases} \varphi_1(t), & t \leq T, \\ \varphi_2(t - T), & t > T. \end{cases}$$

(Lean: `if t ≤ T then φ₁ t else φ₂ (t - T)`.)

Definition 50 (Extend (chain) two gradient flows). Given a flow φ on $[0, T]$ and a flow ψ on $[0, \delta]$ starting at $\varphi(T)$, the piecewise trajectory (Definition 49) forms a gradient flow on $[0, T + \delta]$. The derivative condition holds throughout: it matches the flow equations on $(0, T)$ and on $(T, T + \delta)$, and at the junction point $t = T$, the derivatives agree by the initial condition of ψ .

Definition 51 (Restrict a flow to a smaller time interval). A flow on $[0, S]$ restricts to one on $[0, T]$ for $T \leq S$ by reusing the same trajectory and weakening the domain proof.

Lemma 52 (Gradient map is continuous). *If $f \in C^1$, then ∇f is continuous by the definition of the gradient in terms of the Fréchet derivative.*

Lemma 53 (Norm bound from a Lipschitz gradient). *If ∇f is L -Lipschitz, then by the reverse triangle inequality,*

$$\|\nabla f(x)\| \leq \|\nabla f(y)\| + \|\nabla f(x) - \nabla f(y)\| \leq \|\nabla f(y)\| + L\|x - y\|.$$

Theorem 54 (Local gradient flow existence). *If $f \in C^1$ and ∇f is globally L -Lipschitz, then for any x_0 there exists $\delta > 0$ and a gradient flow on $[0, \delta]$ starting at x_0 . This is a direct application of the Picard–Lindelöf existence theorem for ODEs to the vector field $x \mapsto -\nabla f(x)$.*

Definition 55 (Concrete local flow constructor). A concrete local flow can be constructed explicitly on the time step

$$\delta := \frac{1}{4L + 4}.$$

This formula comes from applying the Picard–Lindelöf theorem with careful tracking of the constants in the problem.

Theorem 56 (Uniform local existence). *There exists $\delta > 0$ (in fact $\delta = 1/(4L + 4)$) such that for every x_0 there is a gradient flow on $[0, \delta]$ starting from x_0 .*

Lemma 57 (Iterated extension). *From the local existence result (Theorem 54) with the uniform time step $\delta = 1/(4L + 4)$, we can construct flows on $[0, n\delta]$ for any $n \in \mathbb{N}$ by repeatedly extending the flow at each time interval $[(k - 1)\delta, k\delta]$.*

Theorem 58 (Gradient flow existence on $[0, T]$). *If $f \in C^1$, ∇f is globally L -Lipschitz, and $T > 0$, then for any x_0 there exists a gradient flow on $[0, T]$ starting at x_0 .*

Proof. We choose $\delta = 1/(4L + 4)$ from Theorem 56. Let $n = \lceil T/\delta \rceil + 1$, so that $T \leq n\delta$. By Lemma 57, we obtain a gradient flow on $[0, n\delta]$. Finally, by Definition 51, we restrict to $[0, T]$ to get the desired flow. $\square$

Definition 59 (Global gradient flow). A global gradient flow for initial condition x_0 is a trajectory $\varphi : \mathbb{R} \rightarrow \mathcal{H}$ with $\varphi(0) = x_0$ and satisfying the ODE $\varphi'(t) = -\nabla f(\varphi(t))$ for all $t \geq 0$.

Definition 60 (Restriction of a global flow). A global flow restricts to a bounded flow on $[0, T]$ by reusing the same trajectory.

Lemma 61 (Uniqueness of bounded flows). *If ∇f is L -Lipschitz and $T \geq 0$, then two gradient flows with the same initial condition must coincide on $[0, T]$. This follows from the standard ODE uniqueness theorem applied to the Lipschitz vector field $x \mapsto -\nabla f(x)$.*

Lemma 62 (Consistency at overlaps). *For a family of flows $\varphi(n)$ on $[0, n + 1]$, any two indexes agree at any time t in both domains. This follows from uniqueness: the two flows satisfy the same ODE with the same initial data, so they must coincide on the overlap.*

Lemma 63 (Local agreement of the glued global trajectory). *Define the global trajectory by $x(t) := (\varphi(\lceil t \rceil)).\text{trajectory}(t)$. For each $t \geq 0$, near t this trajectory agrees with the bounded flow $(\varphi(\lceil t \rceil)).\text{trajectory}$, so the derivative at t transfers from the bounded flow.*

Theorem 64 (Global gradient flow existence for all $t \geq 0$). *If $f \in C^1$ and ∇f is globally L -Lipschitz, then for any x_0 there exists a global gradient flow defined for all $t \geq 0$.*

Proof. For each $n \in \mathbb{N}$, apply Theorem 58 to obtain a bounded gradient flow $\varphi(n)$ on $[0, n + 1]$. Define the global trajectory by $x(t) := (\varphi(\lceil t \rceil)).\text{trajectory}(t)$ for all $t \geq 0$. By Lemma 63, for each $t \geq 0$, the trajectory x agrees locally with $\varphi(\lceil t \rceil)$ near t . Therefore, x satisfies the ODE for all $t \geq 0$ by transfer of derivatives from the local agreement. $\square$

Chapter 7

Convexity

Throughout this section, E is a real normed space, $f : E \rightarrow \mathbb{R}$, $s \subseteq E$ is a convex set, and $x, y \in E$.

Definition 65 (Convex function on a set). We say that f is *convex on s* if s is a convex set and for all $x, y \in s$ and $a, b \geq 0$ with $a + b = 1$:

$$f(ax + by) \leq af(x) + bf(y).$$

Theorem 66 (Local minimizer on convex set is global). *If f is convex on s and $x \in s$ is a local minimizer of f restricted to s , then x is a global minimizer of f on s .*

Theorem 67 (Set of minimizers is convex). *If f is convex on s , then the set of minimizers*

$$\mathcal{M} := \{x \in s \mid \forall y \in s, f(x) \leq f(y)\}$$

is convex.

Proof. Let $x, y \in \mathcal{M}$ and $a, b \geq 0$ with $a + b = 1$. Set $z := ax + by$. Since s is convex, $z \in s$. For any $w \in s$, convexity of f gives

$$f(z) \leq af(x) + bf(y).$$

Since x and y are both minimizers, $f(x) \leq f(w)$ and $f(y) \leq f(w)$, so

$$f(z) \leq af(x) + bf(y) \leq af(w) + bf(w) = (a + b)f(w) = f(w).$$

Hence $z \in \mathcal{M}$. If $\mathcal{M}$ is empty, it is trivially convex. □

Lemma 68 (First-order characterization of convexity). *If f is convex on s and Fréchet differentiable at $x \in s$, then for all $y \in s$:*

$$f(x) + Df(x)(y - x) \leq f(y).$$

Proof. Step 1: Convexity upper bound along the segment. For $\alpha \in [0, 1]$, the convexity of f gives

$$f(x + \alpha(y - x)) = f((1 - \alpha)x + \alpha y) \leq (1 - \alpha)f(x) + \alpha f(y).$$

Step 2: Rearranging into a difference quotient. For $\alpha \in (0, 1]$, subtracting $f(x)$ and dividing by $\alpha > 0$ yields

$$\frac{f(x + \alpha(y - x)) - f(x)}{\alpha} \leq f(y) - f(x).$$

Step 3: Taking the limit. By the chain rule, the map $\alpha \mapsto f(x + \alpha(y - x))$ has derivative $Df(x)(y - x)$ at $\alpha = 0$. Therefore the difference quotients on the left converge to $Df(x)(y - x)$ as $\alpha \rightarrow 0^+$. Passing the limit through the inequality gives

$$Df(x)(y - x) \leq f(y) - f(x),$$

which rearranges to $f(x) + Df(x)(y - x) \leq f(y)$. □

Theorem 69 (Zero derivative implies global minimizer). *If f is convex on s , $x \in s$, and $Df(x) = 0$, then x is a global minimizer of f on s .*

Proof. Apply Lemma 68 with the given x and any $y \in s$:

$$f(x) + Df(x)(y - x) \leq f(y).$$

Since $Df(x) = 0$, the second term vanishes, giving $f(x) \leq f(y)$ for all $y \in s$. □

Chapter 8

Polyak–Łojasiewicz condition and convergence

Lemma 70 (Energy gap derivative and PL differential inequality). *Let $g(t) := f(\varphi(t)) - \inf f$, where φ is a global gradient flow. If $f \in C^1$ and f satisfies the global μ -PL condition, then for every $t \geq 0$,*

$$g'(t) = -\|\nabla f(\varphi(t))\|^2 \quad \text{and} \quad g'(t) \leq -2\mu g(t).$$

Proof. Differentiate $g(t) = f(\varphi(t)) - \inf f$. The constant $\inf f$ contributes no derivative, and the chain rule along the flow gives

$$g'(t) = \frac{d}{dt}f(\varphi(t)) = -\|\nabla f(\varphi(t))\|^2.$$

The global PL inequality gives

$$\|\nabla f(\varphi(t))\|^2 \geq 2\mu(f(\varphi(t)) - \inf f) = 2\mu g(t),$$

so multiplying by -1 yields $g'(t) \leq -2\mu g(t)$. □

Lemma 71 (Exponential decay of the energy gap). *Under the same hypotheses, the energy gap satisfies*

$$f(\varphi(t)) - \inf f \leq (f(x_0) - \inf f) e^{-2\mu t}$$

for all $t \geq 0$.

Proof. Consider the auxiliary function

$$h(t) := (f(\varphi(t)) - \inf f) e^{2\mu t}.$$

Using the previous lemma and the product rule, one finds that $h'(t) \leq 0$ for $t \geq 0$, so h is non-increasing. Therefore $h(t) \leq h(0) = f(x_0) - \inf f$, and multiplying by $e^{-2\mu t}$ gives the stated estimate. □

Theorem 72 (Convergence of the gradient flow under the PL condition). *Assume in addition that f is bounded below. Then along any global gradient flow,*

$$f(\varphi(t)) \rightarrow \inf f \quad \text{as } t \rightarrow \infty.$$

Proof. The global PL estimate gives the upper bound from Lemma 71. Since f is bounded below, we also have $f(\varphi(t)) - \inf f \geq 0$ for all $t \geq 0$. The right-hand side is an exponentially decaying term, hence tends to 0 as $t \rightarrow \infty$. By squeezing, the energy gap converges to 0, which is equivalent to $f(\varphi(t)) \rightarrow \inf f$. $\square$

Chapter 9

Strong Convexity

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, $s \subseteq E$ is a convex set, $m > 0$, and $x, y \in E$.

Definition 73 (Strongly convex function on a set). We say that f is *strongly convex on s with modulus $m > 0$* if s is a convex set and for all $x, y \in s$ and $a, b \geq 0$ with $a + b = 1$:

$$f(ax + by) \leq af(x) + bf(y) - \frac{m}{2} ab \|x - y\|^2.$$

In particular, taking $a = 1 - \alpha$, $b = \alpha$ recovers the modulated convexity inequality

$$f((1 - \alpha)x + \alpha y) \leq (1 - \alpha)f(x) + \alpha f(y) - \frac{m}{2} \alpha(1 - \alpha) \|x - y\|^2.$$

Lemma 74 (Strong convexity reduces to convexity). f is strongly convex on s with modulus m if and only if $x \mapsto f(x) - \frac{m}{2} \|x\|^2$ is convex on s :

$$\text{StrongConvexOn } s \ m \ f \iff \text{ConvexOn } s \left(x \mapsto f(x) - \frac{m}{2} \|x\|^2 \right).$$

Lemma 75 (First-order lower bound for strongly convex functions). If f is strongly convex on s with modulus $m > 0$ and Fréchet differentiable at $x \in s$, then for all $y \in s$:

$$f(y) \geq f(x) + Df(x)(y - x) + \frac{m}{2} \|y - x\|^2.$$

Proof. Define $g : E \rightarrow \mathbb{R}$ by $g(z) = f(z) - \frac{m}{2} \|z\|^2$. By Lemma 74, g is convex on s . Since f is Fréchet differentiable at x and $z \mapsto \frac{m}{2} \|z\|^2$ is differentiable everywhere, g is Fréchet differentiable at x with

$$Dg(x) = Df(x) - D\left(\frac{m}{2} \|\cdot\|^2\right)(x).$$

A direct computation gives $D\left(\frac{m}{2} \|\cdot\|^2\right)(x)(v) = m\langle x, v \rangle$ for all v . Apply Lemma 68 to the convex function g :

$$g(x) + Dg(x)(y - x) \leq g(y).$$

Substituting $g(z) = f(z) - \frac{m}{2} \|z\|^2$ and $Dg(x)(v) = Df(x)(v) - m\langle x, v \rangle$:

$$f(x) - \frac{m}{2} \|x\|^2 + Df(x)(y - x) - m\langle x, y - x \rangle \leq f(y) - \frac{m}{2} \|y\|^2.$$

Rearranging, it remains to show that

$$-\frac{m}{2} \|x\|^2 - m\langle x, y - x \rangle + \frac{m}{2} \|y\|^2 = \frac{m}{2} \|y - x\|^2.$$

Indeed, expanding $\langle x, y - x \rangle = \langle x, y \rangle - \|x\|^2$:

$$\begin{aligned} -\frac{m}{2}\|x\|^2 - m\langle x, y \rangle + m\|x\|^2 + \frac{m}{2}\|y\|^2 &= \frac{m}{2}\|x\|^2 - m\langle x, y \rangle + \frac{m}{2}\|y\|^2 \\ &= \frac{m}{2}(\|x\|^2 - 2\langle x, y \rangle + \|y\|^2) = \frac{m}{2}\|y - x\|^2. \end{aligned}$$

Therefore $f(x) + Df(x)(y - x) + \frac{m}{2}\|y - x\|^2 \leq f(y)$. $\square$

Theorem 76 (Unique global minimizer under strong convexity). *Suppose that f is strongly convex on s with modulus $m > 0$ and Fréchet differentiable at $x^* \in s$. If $Df(x^*) = 0$, then x^* is a global minimizer of f on s , and any other global minimizer $y \in s$ satisfies $y = x^*$.*

Proof. Apply Lemma 75 with $x = x^*$ and any y in the domain:

$$f(y) \geq f(x^*) + Df(x^*)(y - x^*) + \frac{m}{2}\|y - x^*\|^2.$$

Since x^* is a stationary point, $Df(x^*) = 0$, so:

$$f(y) \geq f(x^*) + \frac{m}{2}\|y - x^*\|^2.$$

By the definition of strong convexity, the modulus $m > 0$. Furthermore, for any $y \neq x^*$, the squared norm $\|y - x^*\|^2 > 0$. Therefore, for all $y \neq x^*$, the term $\frac{m}{2}\|y - x^*\|^2$ is strictly positive, which implies:

$$f(y) > f(x^*)$$

This demonstrates that x^* is the strictly unique global minimizer of f . $\square$

Lemma 77 (C^1 regularity of the gradient). *If $f \in C^2$, then $y \mapsto \nabla f(y)$ is in C^1 .*

Lemma 78 (Hessian as Fréchet derivative of the gradient). *If $f \in C^2$, then at every $x \in E$, the map $y \mapsto \nabla f(y)$ has Fréchet derivative $\nabla^2 f(x)$.*

Lemma 79 (Strong convexity $\Rightarrow$ Hessian lower bound). *If $f \in C^2$ is strongly convex on E with modulus m , then*

$$m\|u\|^2 \leq \langle \nabla^2 f(x)u, u \rangle \quad \text{for all } x, u \in E.$$

Proof. Fix x, u and $t > 0$. The first-order strong convexity bound (Lemma 75) at the pair $(x, x + tu)$ and at $(x + tu, x)$ together with summation yields

$$\frac{(Df(x + tu) - Df(x))(u)}{t} \geq m\|u\|^2.$$

As $t \rightarrow 0^+$, the left-hand side converges to $\langle \nabla^2 f(x)u, u \rangle$ (using Lemma 78), giving the claim. $\square$

Lemma 80 (Hessian lower bound $\Rightarrow$ strong convexity). *If $f \in C^2$ and $m\|u\|^2 \leq \langle \nabla^2 f(x)u, u \rangle$ for all $x, u \in E$, then f is strongly convex on E with modulus m .*

Proof. By Lemma 74, it suffices to show that $g(y) := f(y) - \frac{m}{2}\|y\|^2$ is convex on E . For any line restriction $\varphi(t) := g(x + td)$, a direct computation gives $\varphi''(t) = \langle \nabla^2 f(x + td)d, d \rangle - m\|d\|^2 \geq 0$, so φ is convex. Since this holds for every line, g is convex. $\square$

Lemma 81 (Hessian characterization of strong convexity). *Suppose $f \in C^2$. Then f is strongly convex on E with modulus m if and only if $m\|u\|^2 \leq \langle \nabla^2 f(x)u, u \rangle$ for all $x, u \in E$, i.e. $\nabla^2 f(x) \succeq mI$.*

Proof. Direct from Lemmas 79 and 80. $\square$

Lemma 82 (L -smoothness $\Rightarrow$ Hessian upper bound). *If $f \in C^2$ and f is globally L -smooth, then $\langle \nabla^2 f(x) u, u \rangle \leq L \|u\|^2$ for all $x, u \in E$.*

Proof. Apply Theorem 31 with the fact that ∇f has Fréchet derivative $\nabla^2 f(x)$ at each x (Lemma 78). $\square$

Lemma 83 (Operator-norm bound for symmetric PSD operators with quadratic-form bound). *Let $A : E \rightarrow_L E$ be self-adjoint with $0 \leq \langle Au, u \rangle$ and $\langle Au, u \rangle \leq c \|u\|^2$ for all u , where $0 \leq c$. Then $\|A\| \leq c$.*

Proof. By the polarization identity (using self-adjointness),

$$4 \langle Au, v \rangle = \langle A(u+v), u+v \rangle - \langle A(u-v), u-v \rangle.$$

Combining with $0 \leq \langle Aw, w \rangle \leq c \|w\|^2$ gives $|\langle Au, v \rangle| \leq \frac{c}{2} (\|u\|^2 + \|v\|^2)$. Substituting $v = (\|u\|/\|Au\|) Au$ yields $\|Au\| \leq c \|u\|$, hence $\|A\| \leq c$. $\square$

Lemma 84 (Self-adjointness of the Hessian). *If $f \in C^2$, then for every $x, u, v \in E$:*

$$\langle \nabla^2 f(x) u, v \rangle = \langle u, \nabla^2 f(x) v \rangle.$$

Proof. By the symmetry of the second Fréchet derivative (`ContDiffAt.isSymmSndFDerivAt`), $(D^2 f(x))(u)(v) = (D^2 f(x))(v)(u)$. Combining with the Riesz duality used to define the gradient (∇f is the composition of Df with $(\text{InnerProductSpace.toDual})^{-1}$) and the chain rule transfers the symmetry to the Hessian. $\square$

Lemma 85 (Hessian upper bound + convexity $\Rightarrow L$ -smoothness). *Suppose $f \in C^2$ is convex on E and $\langle \nabla^2 f(x) u, u \rangle \leq L \|u\|^2$ for all x, u . Then f is globally L -smooth.*

Proof. By Theorem 30, it suffices to bound $\|\nabla^2 f(x)\| \leq L$ for all x . The Hessian is self-adjoint by Lemma 84. Since f is convex, it is strongly convex with modulus 0, so Lemma 79 gives $\langle \nabla^2 f(x) u, u \rangle \geq 0$. Conclude by Lemma 83. $\square$

Corollary 86 (Hessian sandwich characterization of L -smooth convex functions). *Suppose $f \in C^2$ is convex on E . Then f is globally L -smooth if and only if $\langle \nabla^2 f(x) u, u \rangle \leq L \|u\|^2$ for all $x, u \in E$, i.e. $\nabla^2 f(x) \preceq LI$.*

Proof. Direct from Lemmas 82 and 85. $\square$

Lemma 87 (Inner-product expansion of $\|a + r b\|^2$). *For $a, b \in E$ and $r \in \mathbb{R}$:*

$$\|a + r b\|^2 = \|a\|^2 + 2r \langle a, b \rangle + r^2 \|b\|^2.$$

Theorem 88 (Strong convexity $\Rightarrow$ global PL). *If f is strongly convex on E with modulus $m > 0$ and Fréchet differentiable on E , then f is globally m -PL:*

$$\|\nabla f(x)\|^2 \geq 2m (f(x) - \inf f) \quad \text{for all } x \in E.$$

Proof. Fix x and set $g := \nabla f(x)$. By the first-order lower bound (Lemma 75), for every $z \in E$,

$$f(z) \geq f(x) + \langle g, z - x \rangle + \frac{m}{2} \|z - x\|^2.$$

Expanding $0 \leq \|g + m(z - x)\|^2$ via Lemma 87 gives

$$0 \leq \|g\|^2 + 2m \langle g, z - x \rangle + m^2 \|z - x\|^2.$$

Multiplying the first inequality by $2m > 0$ and adding to the second yields

$$f(x) - \frac{1}{2m} \|g\|^2 \leq f(z) \quad \text{for all } z \in E.$$

Hence f is bounded below and $\inf f \geq f(x) - \frac{1}{2m} \|g\|^2$, which rearranges to $\|g\|^2 \geq 2m(f(x) - \inf f)$. $\square$